Deterministic Identity Testing for Constant-Width Affine ROABP Orbits in Every Characteristic

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Abstract

We give a self-contained candidate proof of a polynomial-size black-box hitting set, for every fixed width, for the unknown invertible affine orbit of read-once oblivious algebraic branching programs in arbitrary characteristic. The local univariate matrices may have arbitrary generic ranks and arbitrary degrees; only the scalar total degree is bounded. The proof extends a coefficient-span argument to finite monomial Artin rings. A box quotient then bounds the support of every monomial of a lowest homogeneous component whenever all interval ranks are preserved. The standard characteristic-order criterion for Hasse pivots bounds every exponent of each such leading monomial. Conditioning on the other blocks supplies the required filtered interface in an exterior-power rank descent. The resulting explicit map is a moment curve plus $K(w) = 2^{O(w^2)}$ Shpilka–Volkovich pairs. Its basic grid has at most $(nD+1)^{2K(w)+1}$ points for scalar total degree at most D.

Status and attribution. The constituent notes and this self-contained integrated manuscript have separate model-assisted internal reconstructions and exact-source audits, including an additional adversarial review of the Artin reset/source step. The mathematical source is frozen after those internal reviews. It remains an internal candidate proof, not external peer review or a publication-priority claim. The generic Hasse-order closure used below is a standard p-adic criterion, recorded for example in Garcia–Lax [2, Proposition 1.1] and in the Stöhr–Voloch order-sequence context [3]; no novelty is claimed for that criterion. We prove the precise rational-vector version needed here rather than importing extra geometric hypotheses.

1 Statement and oracle model

Let $n, w, D \geq 1$. Suppose that the nonzero polynomial presented to the oracle has the form

$$f(X) = uA_1(\ell_1(X) + c_1) \cdots A_n(\ell_n(X) + c_n)v, \qquad \deg f \leq D. \tag{1}$$

The matrices A_i are w-by-w univariate polynomial matrices over a field $\mathbb{F}$, and u, v are constant endpoints. Their generic ranks and entry degrees are unrestricted. The forms $\ell_1, \ldots, \ell_n$ are the coordinates of an invertible linear transformation. The representation, its order, and the affine transformation are unknown. A representation of smaller width can be padded with zeros.

This is the affine-orbit model for one unknown ROABP order, as studied by Bhargava and Ghosh [1]; it does not assume a representation in every variable order. The degree promise is on the scalar polynomial. If local entry degrees, or latent scalar individual degrees, are bounded by δ, one may take $D = n\delta$. No truncation of the promised representation is performed.

For $w \geq 2$ and $1 \leq r < w$ define

$$W_{w,r} = \binom{w}{r}, \tag{2}$$

$$h_{w,r} = (r+2)\prod_{s=r+2}^{w}\left(\binom{s-1}{r-1}+1\right)-2, \tag{3}$$

$$k_{w,r} = (W_{w,r}^2-1)(2h_{w,r}+2), \qquad K(w)=\sum_{r=1}^{w-1} k_{w,r}. \tag{4}$$

An empty product is one, and $K(1)=0$. These constants are independent of the characteristic and of every degree parameter. The construction gives

w	1	2	3	4	5	6
$K(w)$	0	12	128	1500	23718	553854

Choose n distinct nodes a_i in a working field and their Lagrange polynomials $\lambda_i(y)$, of degree at most $n-1$. Write

$$\mathrm{SV}_{k,i}(z,y)=\sum_{j=1}^{k} z_j\lambda_i(y_j), \qquad \gamma_i(t)=t^i.$$

Theorem 1.1. *In every characteristic, every nonzero polynomial (1) satisfies*

$$f\big(\gamma(t)+\mathrm{SV}_{K(w)}(z,y)\big)\neq 0. \tag{5}$$

The pure map $\mathrm{SV}_{K(w)+1}$ also hits the class. A Cartesian grid for (5) uses at most

$$(nD+1)(D+1)^{K(w)}\big((n-1)D+1\big)^{K(w)} \le (nD+1)^{2K(w)+1} \tag{6}$$

oracle evaluations. A working field of size at least $nD+1$ supplies all grids and nodes; extension-field oracle evaluations are permitted when needed. Moreover

$$K(w)\le (w-1)2^{w(w+1)/2+2w}=2^{O(w^2)} \qquad (w\ge 2).$$

Thus for every fixed width the hitting set is polynomial in n, D. Constants, including the cases $n=0$ or $D=0$, need one query.

This is an XP-type query bound in the width parameter: the polynomial exponent depends on w. It does not establish an FPT bound $g(w)(nD)^{O(1)}$ with a uniform exponent. The parameter dependence is large and is not asserted to be optimal. The theorem concerns invertible affine transformations, not singular projections or orbit closures. All final degree counts refer to the original scalar f, even though auxiliary minors may have much larger degrees.

2 Coefficient spaces in monomial Artin rings

Let E be a field. For disjoint variable blocks X_i, let $\mathcal{R}_i = E[X_i]/I_i$, where I_i is a proper monomial ideal containing a positive power of every variable. A trivial block $\mathcal{R}_i = E$ is also allowed. The surviving monomials form an E-basis, and the augmentation ideal is nilpotent. Work in the tensor product $\mathcal{R}=\bigotimes_i \mathcal{R}_i$ over E. Support counts the blocks containing a nonconstant monomial.

For a row-valued or matrix-valued polynomial over these rings, Coeff means the E-linear span of its actual coefficient rows in the surviving monomial bases.

Lemma 2.1. *Scalar multiplication, including multiplication by a polynomial using future variable blocks, only takes E-linear combinations of existing coefficient rows. The same holds for a compatible matrix multiplying a row family on the left. Left multiplication by a square matrix with invertible constant term preserves the coefficient-row span in both directions.*

Proof. Coefficient convolution expresses each resulting row as a linear combination of original rows. Monomial truncation discards terms and never identifies a higher monomial with a lower one. A matrix with invertible constant term has an inverse in the finite local ring. Apply the inclusion to the matrix and its inverse. □

Definition 2.2. *An interval matrix has* unit rank s *if its constant matrix has field rank s and all its $(s+1)$-minors vanish in the tensor ring. At $s = 0$ this means that the matrix is zero; at full width the condition on larger minors is empty.*

The constant matrix supplies an s-minor with nonzero constant determinant, which is a unit. Thus the definition imposes the precise algebraic rank identities needed below without defining a fraction-field rank over a ring with zero divisors.

Lemma 2.3. *Let P have unit rank s, and suppose an s-by-s block E_0 has invertible constant term. After arranging it in the upper left, write $P = \left[\begin{smallmatrix} E_0 & B \\ C & D \end{smallmatrix}\right]$. Then $D = CE_0^{-1}B$. Consequently every row in the row module of P is determined by its first s coordinates; a zero active part forces the entire row to be zero. Constant invertible cutwise gauges preserve all unit-rank conditions.*

Proof. Each entry of the Schur complement $D - CE_0^{-1}B$ is, up to sign, an $(s+1)$-minor divided by the unit $\det E_0$, and hence vanishes. The row-graph statement follows from $P = \left[\begin{smallmatrix} I \\ CE_0^{-1} \end{smallmatrix}\right] E_0[I, E_0^{-1}B]$. Cauchy–Binet preserves the vanishing larger minors under constant invertible changes, and the constant rank is unchanged. □

No integral-domain assumption is present in these lemmas.

3 Weighted concentration with a rank palette

Fix a width W and a set of allowed ranks

$$\mathcal{P} = \{0 = p_0 < p_1 < \cdots < p_m = W\}.$$

Assume every contiguous interval of a word $A_1(X_1)\cdots A_n(X_n)$ over the tensor rings has unit rank equal to its constant rank, and that rank belongs to $\mathcal{P}$. Unused ranks may be included in the palette.

For $a = p_j \in \mathcal{P}$ put

$$G(a) + 1 = \prod_{\ell=j+1}^{m} (p_\ell - p_{\ell-1} + 1). \tag{7}$$

In particular $G(W) = 0$. For $b > a$ in the palette,

$$(b - a + 1)(G(b) + 1) \le G(a) + 1. \tag{8}$$

Indeed inserting an intermediate level splits a jump into positive u, v and replaces $u + v + 1$ by $(u+1)(v+1) \ge u + v + 1$.

Suppose constant cutwise gauges have made

$$A_i(0) = \operatorname{diag}(I_a, Q_i), \qquad a \in \mathcal{P}.$$

Let the initial row space be

$$S_0 = \operatorname{rowspan}(B_0) \oplus (0 \times K_0),$$

where the active part H of B_0 has independent rows and spans a space T. Assign cost zero to rows of B_0 and cost one to rows $(0, p)$ with $p \in K_0$. A weighted budget b allows coefficients of $B_0 A_1 \cdots A_n$ supported on at most b blocks and coefficients from the initial passive rows supported on at most $b - 1$ blocks. Assume that the active projection of the full reachable coefficient-row space S_i remains exactly T at every cut.

Lemma 3.1 (Weighted stable phase)**.** *Under these assumptions, weighted budget $G(a)$ spans S_n.*

Proof. Induct on the number of palette levels above a. At $a = W$ there are no passive coordinates. The unchanged active space is already spanned by the constant-prefix rows, and $G(W) = 0$ suffices. We give the exact kernel and reset identities before the induction step.

Kernel dynamics. Propagate $B_i = B_{i-1}A_i(0)$; its active part remains H. Write $S_i = \mathrm{rowspan}(B_i) \oplus (0 \times K_i)$ and put $k = W - a$. For $d = \dim T > 0$, choose a field right inverse J of H. The active block of each local matrix is a unit, so write

$$A_i = \begin{bmatrix} E_i & E_iC_i \\ D_iE_i & D_iE_iC_i + F_i \end{bmatrix}, \qquad E_i(0) = I_a, \quad C_i(0) = D_i(0) = 0, \quad F_i(0) = Q_i.$$

Active stability gives

$$(H + B_{i-1,p}D_i)E_i = R_iH, \qquad R_i(0) = I_d,$$

where the subscript p denotes the passive block. Define

$$W_i = R_i^{-1}B_{i-1}A_i = [H, B_{i,p} + Z_i], \tag{9}$$

$$\widetilde{F}_i = (I_k - D_iE_iJR_i^{-1}B_{i-1,p})F_i. \tag{10}$$

These functions use only the current variable block: B_{i-1}, H, J are field-constant coefficient lifts. Moreover $Z_i(0) = 0$ and $\widetilde{F}_i(0) = Q_i$. Lemma 2.1 gives $\mathrm{Coeff}(W_i) = \mathrm{Coeff}(B_{i-1}A_i)$. Each coefficient row of Z_i is consequently a combination of actual rows of weighted cost at most one.

For $p \in K_{i-1}$, stability implies $pD_iE_iJH = pD_iE_i$. Direct subtraction gives

$$(0, p)A_i - (pD_iE_iJ)W_i = (0, p\widetilde{F}_i).$$

Taking coefficient spans in both directions, and subtracting active parts using the available constant rows B_i, yields

$$K_i = \mathrm{Coeff}(Z_i) + \mathrm{Coeff}(K_{i-1}\widetilde{F}_i). \tag{11}$$

For a reachable state $a_{i-1}B_{i-1} + (0, p_{i-1})$, its coordinates evolve by the exact identities

$$a_i = a_{i-1}R_i + p_{i-1}D_iE_iJ, \qquad p_i = p_{i-1}\widetilde{F}_i + a_iZ_i. \tag{12}$$

If $d = 0$, the B, W, Z families are empty; active stability forces reachable active outputs to zero, and the formulas reduce to $K_i = \mathrm{Coeff}(K_{i-1}F_i)$. No right inverse is required in that case.

Reset across a zero passive interval. Suppose $Q_c \cdots Q_e = 0$. The original interval $P = A_c \cdots A_e$ has unit rank a, and its active pivot is a unit. It is a graph over the active coordinates by Lemma 2.3. For fixed $p \in K_{c-1}$, the active part of $B_{c-1}P$ is $R_{[c,e]}H$ with $R_{[c,e]}(0) = I$, and that of $(0, p)P$ is coefficientwise in T. A regular initial active control a_{c-1} therefore makes

$$(a_{c-1}B_{c-1} + (0, p))P$$

have zero final active part. The graph identity makes its entire output zero. Since H has a constant right inverse, its final active coordinate a_e is zero as well. Unrolling (12) gives

$$p\widetilde{F}_c \cdots \widetilde{F}_e = -\sum_{j=c}^{e-1} a_jZ_j\widetilde{F}_{j+1} \cdots \widetilde{F}_e. \tag{13}$$

The $j = e$ term is absent because $a_e = 0$. When $d = \dim T = 0$, the reachable row already has zero active part and the same reset statement holds without controls or sources.

The initial control may use future interval variables. This does not violate reachability: by Lemma 2.1, multiplying an actual prefix row by such a scalar only combines its coefficient rows. Every controlled intermediate state remains coefficientwise in S_i, and its passive remainder remains in K_i. The reset identity is used only for an *unweighted coefficient-span inclusion*. No support cost is assigned to its possibly complicated scalar multipliers.

Positive passive constant rank. Suppose $q = \operatorname{rank}(Q_1 \cdots Q_n) > 0$, and put $b = a + q$. This is the rank of the full constant interval, so $b \in \mathcal{P}$. Constant gauges split a persistent q-dimensional passive core. For explicit construction, choose U_0 complementary to the kernel of $Q_1 \cdots Q_n$, set $U_i = U_0 Q_1 \cdots Q_i$, choose a complement V_n of U_n, and put

$$V_i = \{v : vQ_{i+1} \cdots Q_n \in V_n\}.$$

Then U_i, V_i are complementary of dimensions $q, k-q$; each constant map preserves this splitting, and maps the U_i isomorphically. The constants become $\operatorname{diag}(I_a, I_q, Q_i')$.

Let β be the dimension of the initial passive kernel's projection onto this new core. The enlarged reachable active space starts at dimension $\dim T + \beta$, is nested, and has dimension at most $\dim T + q$. Hence there are at most $q - \beta$ strict growth steps. Every stable subphase now has persistent core dimension b, so induction applies with budget $G(b)$.

Here is the support accounting. If active lifts cost at most L and passive-kernel generators at most $L+1$, a stable phase costs at most $L+G(b)$. The next growth step adds one fresh block; a new active basis and its kernel can both be selected at cost $L + G(b) + 1$. Substituting these coefficient witnesses is legitimate because later blocks are disjoint from earlier ones. The tensor quotient changes no cross-block coefficient convolution.

If $\beta = 0$, the initial costs are zero and one. At most q growth steps give

$$(q+1)G(b) + q = (q+1)(G(b)+1) - 1 \le G(a)$$

by (8). If $\beta \ge 1$, choose additional active lifts from the initial kernel at cost one. The remaining kernel also costs one, within the permitted cost two. At most $q - \beta$ growths give

$$1 + (q - \beta + 1)G(b) + (q - \beta) = (q - \beta + 1)(G(b) + 1) \le q(G(b) + 1) \le G(a).$$

The last inequality uses $G(b) \ge 0$ and (8). Only larger active levels $b > a$ have been used.

Zero passive constant rank. Suppose $Q_1 \cdots Q_n = 0$. Choose c maximal with $Q_c \cdots Q_n = 0$. Every proper suffix $Q_{j+1} \cdots Q_n$, $j \ge c$, has positive rank, including the empty suffix of rank k. Equations (11) and (13) show that K_n is spanned by coefficients of

$$Z_j \widetilde{F}_{j+1} \cdots \widetilde{F}_n, \qquad j \ge c.$$

For a coefficient row p of Z_j, start the actual suffix $A_{j+1} \cdots A_n$ from $\operatorname{span}(B_j, (0, p))$, assigning weights zero and one. Its old active projection remains T: it is contained in the global reachable space, and its constant-prefix rows already supply T. Its positive-passive-rank case, just proved, has budget $G(a)$. Its kernel recurrence uses the same $B_i, W_i, Z_i, \widetilde{F}_i$ and contains $p\widetilde{F}_{j+1} \cdots \widetilde{F}_n$.

That response is therefore spanned by actual suffix coefficients from B_j of support at most $G(a)$, or from p of support at most $G(a) - 1$. In the original word B_j costs zero and p costs at most one. Its source cost is already paid by the weight one, so the total budget is $G(a)$. This is not circular induction at level a: the positive-rank case invoked only levels $b > a$. Adding the constant rows B_n completes the span of S_n.

Preservation of the palette. Every subphase and source restart uses a contiguous subword of the same full matrices. It changes an initial row space, not the matrices. The cutwise gauges are invertible field-constant changes and preserve each interval rank and its unit-pivot identities. Each newly enlarged core dimension is the rank of an actual full constant interval, hence belongs to $\mathcal{P}$. The corrected passive operators $\widetilde{F}_i$ are used only in the exact algebra; no rank palette is assumed for their products. These observations justify all induction calls and finish the proof. □

Theorem 3.2 (Palette row concentration)**.** *For every fixed row u, the coefficient-row span of $uA_1 \cdots A_n$ is generated by actual coefficients supported on at most*

$$h(\mathcal{P}) = \prod_{j=1}^{m} (p_j - p_{j-1} + 1) - 2 \tag{14}$$

blocks.

Proof. Use Lemma 3.1 with empty active core $a = 0$ and initial passive space $\mathrm{span}(u)$ of weight one. Weighted budget $G(0)$ means ordinary support budget $G(0) - 1$, which is (14). A full interval of unit rank zero is the zero matrix and causes no exception. □

For the full palette $\{0, 1, \ldots, W\}$, this gives $2^W - 2$. The palette refinement will be substantially smaller on compound words.

4 Box quotients bound every leading support

Return to polynomial or regular rational univariate matrices $A_i(Y_i)$ over a field E. Suppose every contiguous interval has generic rank equal to its constant rank at zero, and all these ranks belong to $\mathcal{P}$. Let u, v be any fixed vectors over E and assume $g(Y) = uA_1(Y_1) \cdots A_n(Y_n)v \neq 0$.

Theorem 4.1. *Every nonzero monomial in the first ordinary-homogeneous component of g has support at most $h(\mathcal{P})$.*

Proof. A nonzero constant component is immediate. Otherwise let $m > 0$ be the first degree and choose any nonzero coefficient g_α with $|\alpha| = m$. Set variables outside its support to zero and pass to the tensor box

$$\bigotimes_{i:\alpha_i>0} E[Y_i]/(Y_i^{\alpha_i+1}).$$

Every surviving monomial is a divisor of Y^α. Each proper divisor has total degree below m and zero scalar coefficient. Thus the entire scalar in this quotient is exactly $g_\alpha Y^\alpha$, which is nonzero.

All interval hypotheses survive the quotient. Larger minors vanish identically before quotienting, while every chosen constant nonzero pivot stays a unit. Regular rational denominators are units as well. Theorem 3.2 therefore applies in this particular box. If $|\operatorname{supp} \alpha| > h(\mathcal{P})$, every scalar coefficient permitted by its row-span bound is zero, forcing the entire scalar to be zero. This contradicts $g_\alpha \neq 0$. □

The box step is essential. Ordinary unfiltered coefficient concentration alone would not bound every leading support: it could use a low-support coefficient of a much higher total degree. The quotient removes those coefficients while retaining all needed Schur identities. The endpoints here are arbitrary constants over E; they may depend on other shifts or parameters, as long as they are fixed in the residual variables being considered.

5 Standard Hasse orders and the filtered exponent bound

In characteristic $p > 0$, let $\mathrm{wt}_p(a)$ be the sum of the base-p digits of a. The digit degree of a monomial is the sum of the digit weights of its exponents. In characteristic zero use $\mathrm{wt}_0(a) = a$. For a rational function, Hasse derivatives are defined by

$$F(T+Z) = \sum_{j \geq 0} D_T^{[j]} F(T) Z^j.$$

Their product and composition rules hold over every field.

Lemma 5.1 (Generic pivot closure)**.** *Let $F(T)$ be a nonzero row vector of q rational functions over an arbitrary original field F_0. Call ν a pivot if $D^{[\nu]}F$ does not lie in the $F_0(T)$-span of all earlier jets. There are at most q pivots. If ν is a pivot and $\binom{\nu}{\mu} \neq 0$ in F_0, with $0 \leq \mu \leq \nu$, then μ is a pivot. Consequently every pivot satisfies*

$$\mathrm{wt}_p(\nu) \leq q - 1$$

in positive characteristic, and $\nu \leq q - 1$ in characteristic zero.

Proof. If μ were not a pivot, write

$$D^{[\mu]}F = \sum_{j<\mu} a_j(T) D^{[j]}F.$$

Apply $D^{[\nu-\mu]}$. The left side is $\binom{\nu}{\mu} D^{[\nu]}F$. The product rule on the right gives only jets of indices $j + \nu - \mu - e < \nu$, for $e \geq 0$. This contradicts the pivot property of ν.

In characteristic p, the nonzero binomial coefficients are exactly the digit subindices, as follows from the Frobenius expansion of $(1+Z)^\nu$. If $\nu = \sum_e \nu_e p^e$, all $\prod_e (\nu_e + 1)$ digit subindices are pivots. Thus

$$\mathrm{wt}_p(\nu) + 1 \leq \prod_e (\nu_e + 1) \leq q.$$

In characteristic zero every index $0, \ldots, \nu$ is a pivot. This is the standard p-adic order criterion cited above, proved here for the precise rational-vector setting. □

Lemma 5.2. *Let $C \in E$ be transcendental over F_0, and let $\Lambda : E^q \to E$ be any E-linear functional. If $\Lambda(F(C+Y))$ is nonzero and first becomes nonzero at order a, then a is a generic pivot of F. In particular $\mathrm{wt}_p(a) \leq q - 1$, with ordinary weight in characteristic zero.*

Proof. Every original-field rational jet relation is defined at C because its nonzero denominator cannot vanish there. Every nonzero finite jet minor remains nonzero as well, so all successive jet ranks and pivots are preserved at C. If a were not a pivot, its specialized jet would be a linear combination of earlier ones, all annihilated by Λ, contradicting its nonzero value. □

The functional may depend on C and other correlated parameters. Neither perfection of F_0 nor transcendence of C over the enlarged field E is assumed.

Lemma 5.3 (Every leading exponent)**.** *Let $A_i(T_i)$ be rational univariate matrices of width W over F_0, and let every $C_i \in E$ be transcendental over F_0. For arbitrary fixed E-endpoints, suppose*

$$g(Y) = uA_1(C_1 + Y_1) \cdots A_n(C_n + Y_n)v \neq 0.$$

For every nonzero monomial Y^α of its first homogeneous component, every coordinate satisfies $\mathrm{wt}_p(\alpha_i) \leq W^2 - 1$. Local ranks are unrestricted.

Proof. Let $m = |\alpha|$. Hold all exponents other than that of Y_i at their chosen values. The remaining scalar coefficient at order a is an E-linear matrix-entry functional of $D^{[a]}A_i(C_i)$: its left and right vectors are the products of the fixed outside coefficient matrices and endpoints. At every $a < \alpha_i$ this coefficient is zero, since its full total degree is below m. At $a = \alpha_i$ it is nonzero. Apply Lemma 5.2 to the vector of the W^2 original matrix entries. Constants and empty variable blocks have only order-zero coefficients and cause no exception. □

This is an ordinary-order filtration argument for every leading monomial. It is stronger than selecting a low-digit-weight coefficient somewhere in the expansion. Only individual original-field transcendence is needed; there is no derivative-separating or common-translation premise in Lemma 5.3.

6 Affine detection of a bounded leading component

Lemma 6.1. *Let a nonzero polynomial or regular series have a first homogeneous component of digit degree at most k. After any invertible linear residual change of coordinates, SV_k detects that component and the whole series remains nonzero under the substitution.*

Proof. An invertible linear change preserves nonzeroness. Digit degree cannot increase: expand each exponent into its digit copies, use Frobenius on every copy, and note that collisions can only decrease digit weight. In characteristic zero this is ordinary total degree preservation. Every resulting monomial has support at most k.

The image of SV_k contains every coordinate subspace of dimension at most k: specialize its direction variables to the corresponding interpolation nodes and unused amplitudes to zero. Restricting to the support of a nonzero monomial retains that coefficient, so one of these subspaces has a nonzero restriction. Thus the first component's SV image is nonzero. Higher homogeneous components have larger total degree in the fresh amplitudes and cannot cancel it. A nonzero constant component needs no fresh variables. □

All statements remain valid over a coefficient extension generated by earlier symbolic seeds; evaluation at the interpolation nodes still gives the identity matrix for the Lagrange basis.

7 The interval-rank descent

Adjoin the constant width-w factors

$$E_{\mathrm{in}} = e_1 u, \qquad E_{\mathrm{out}} = v e_1^T$$

to the original latent word. Its full product is $f_{\mathrm{latent}}(T)E_{11}$. These are empty variable blocks and add no observed coordinates. Every interval mentioned in the descent belongs to this *unshifted augmented word*.

At any stage let G be the moment curve plus already introduced SV pairs, with all seeds retained as independent indeterminates. Put

$$C_i = c_i + \ell_i(G).$$

Every C_i is transcendental over the original working field: its t-part is a nonzero linear combination of $t, t^2, \ldots, t^n$, whereas all additional SV terms are independent of t. This remains true in every characteristic, even if an ordinary derivative of that combination vanishes.

Before the rank-r step the induction invariant is

$$\begin{gathered}\text{every augmented interval of generic rank greater than } r \\ \text{retains that rank at the current } C.\end{gathered} \tag{15}$$

At $G = \gamma$, every rank-w interval consists solely of full-rank locals. Their nonzero original-field univariate determinants survive at the individually transcendental C_i, so their product does too. Constant invertible factors also survive. This starts the induction at $r = w - 1$. Previously preserved minors stay nonzero after adding fresh pairs, because setting their amplitudes to zero recovers the old shift.

7.1 Greedy blocks inside one target interval

Fix a contiguous interval I of the unshifted augmented word with generic rank exactly r. Every subinterval of I has rank at least r, since the rank of a full product is at most that of any contiguous subproduct. Starting at the left, take a shortest prefix of rank r, then repeat on the remaining suffix as long as such a prefix exists. Write

$$I = B_1 \cdots B_q T, \qquad B_j = C_j A_{i_j}, \tag{16}$$

where A_{i_j} is the last single factor of that shortest prefix. There is at least one block. The chunks C_j and leftover tail T have generic rank greater than r, or are empty. Indeed a shortest prefix cannot reach below r, and all its proper prefixes have higher rank; stopping means the remaining full suffix has higher rank. Every subinterval within one of these chunks has higher rank as well.

The following words all have generic rank exactly r:

$$B_j, \qquad B_j B_{j+1} \ (j < q), \qquad B_q T. \tag{17}$$

They are subintervals of I, giving rank at least r, and each contains a rank-r block, giving rank at most r. Choose for each one a nonzero r-minor. The algorithm does not need to discover this choice; it is used only in the proof.

7.2 Exterior chunks with a preserved palette

Put $W = \binom{w}{r}$ and replace each local matrix by its r-th compound matrix. Cauchy–Binet gives

$$\bigwedge^{r}(AB) = (\bigwedge^{r} A)(\bigwedge^{r} B)$$

in every characteristic. A matrix of rank s has compound rank $\binom{s}{r}$, by reducing it over its field to a diagonal rank-s matrix. Thus each selected minor in (17) is a nonzero scalar entry of a width-W compound word whose locals are still original-field univariate polynomial matrices.

By (15), every interval inside C_j or T has preserved original rank $s > r$. Its compound interval has preserved rank $\binom{s}{r}$. Hence every prepared compound chunk satisfies Theorem 4.1 with palette

$$\mathcal{P}_{w,r} = \left\{0, \binom{r+1}{r}, \binom{r+2}{r}, \ldots, \binom{w}{r}\right\}.$$

Its first positive rank is $r + 1$, and later gaps are $\binom{s-1}{r-1}$. Formula (14) therefore gives exactly $h(\mathcal{P}_{w,r}) = h_{w,r}$ from (3). An empty chunk is the identity and contributes support zero.

7.3 Conditioning the leading monomial of a test

Let $g(Y)$ be one selected nonzero compound-entry test after the current shift C, in independent latent residual variables. It remains a nonzero polynomial: translation by C is an automorphism over the old-seed field. If its constant term is nonzero, it is already preserved. Otherwise choose *any* nonzero monomial Y^{α} of its first degree m.

Fix the exponents outside one prepared chunk. Coefficient extraction there leaves a scalar contraction of the chunk with a fixed row and column over the old-seed field. These endpoints are products of the fixed outside local coefficient matrices; they can depend on those exponents and all shifts, but not on the chunk's residual variables. The chosen chunk coefficient is nonzero. Every coefficient of smaller *total chunk degree* is zero, since it would yield a nonzero coefficient of the full test below degree m with the same outside exponents. The chunk part of α therefore belongs to its first homogeneous scalar component.

Theorem 4.1 bounds its support by $h_{w,r}$. This is valid for arbitrary coefficient-field endpoints, and also when the chunk degree is zero. Each word in (17) contains at most two prepared chunks and at most two singleton layers:

$$C_j A_{i_j}, \qquad C_j A_{i_j} C_{j+1} A_{i_{j+1}}, \qquad C_q A_{i_q} T.$$

Thus every leading monomial of every test has support at most $2h_{w,r} + 2$.

Apply Lemma 5.3 independently to the whole compound test. Its local matrices have width W, are over the original field, and have individually transcendental shifts. Every leading exponent has digit weight at most $W^2 - 1$, including those at singleton layers where no rank-preservation premise has been used. Consequently the *entire* first homogeneous component of each test has digit degree at most

$$(W^2 - 1)(2h_{w,r} + 2) = k_{w,r}.$$

No multivariate boundary vector has been factored. Both support and exponent bounds are applied to first ordinary degrees, rather than to unfiltered coefficient witnesses.

7.4 One stage preserves rank r

Add a fresh $\mathrm{SV}_{k_{w,r}}$ in the observed coordinates. The latent residual change is the original invertible linear transformation, so Lemma 6.1 preserves every selected test. The same map works simultaneously for all tests in all target intervals: they form a finite family of nonzero rational functions in the common old and fresh seed field. No seed specialization is made during the symbolic induction.

Every word in (17) now has specialized rank at least r because its chosen minor survives, and at most r because its larger minors vanish identically. These ranks are therefore exactly r. The following elementary chain fact completes the step.

Lemma 7.1. *If $B_1, \dots, B_q$ have rank r, every adjacent product has rank r, and $B_q T$ has rank r, then $B_1 \cdots B_q T$ has rank r.*

Proof. Inductively the prefix product through B_j has rank r and the same row space as B_j. Multiplication by B_{j+1} has rank r because $B_j B_{j+1}$ does; the new row space equals that of B_{j+1}. At the last block it therefore has the same row space as B_q, whose multiplication by T has rank r. □

The proof uses the tail test $B_q T$ separately, so there is no uncounted three-chunk test after absorbing the tail. We have established (15) for the next lower rank while preserving all previous ranks. Descend through $r = w - 1, \dots, 1$. The final map is the sum of the moment curve and

exactly $K(w)$ fresh SV pairs. The full augmented interval has generic rank one because $f \neq 0$; its rank preservation gives $f(G) \neq 0$. If $w = 1$, the initial curve already preserves the product of nonzero univariate scalars.

8 Quantitative bounds and the pure map

For $1 \leq r < w$, we have $r + 2 \leq 2^{r+1}$ and $\binom{s-1}{r-1} + 1 \leq 2^s$. Thus

$$h_{w,r} + 2 \leq 2^{\sum_{s=r+1}^{w} s} \leq 2^{w(w+1)/2-1}.$$

Since $W_{w,r} \leq 2^w$, the safe estimate

$$K(w) \leq (w-1)2^{w(w+1)/2+2w}$$

follows. Much smaller constants are possible in several small-width special cases; no such refinements are dependencies here.

The observed polynomial after the final substitution has individual seed degrees at most nD in t, D in each amplitude, and $(n-1)D$ in each direction. Cartesian interpolation proves (6). It evaluates only the original scalar polynomial on an explicit polynomial map. Numerical points need not avoid the auxiliary pivots used in the proof, and the degrees of auxiliary minors do not enter this grid calculation.

For the pure map $\mathrm{SV}_{K(w)+1}$, specialize its first pair to $z_0 = t, y_0 = t$ in every characteristic. A nonzero latent combination of this first pair is $c + tq(t)$, where q is a nonzero combination of the independent Lagrange polynomials. It is nonconstant, hence transcendental over the original field. All later SV stages are independent of t. Therefore the entire preceding proof applies to this first-pair specialization, and the unspecialized pure SV image is nonzero as well. A basic pure-map grid uses at most

$$\big((D+1)((n-1)D+1)\big)^{K(w)+1} \leq (nD+1)^{2K(w)+2}$$

points. This completes Theorem 1.1.

9 What the proof does and does not use

The rank profile is used through constant-unit pivots and vanishing larger minors, so it survives monomial box quotients. The weighted reset identity is a coefficient-span statement; it does not claim preservation of a matrix image variety or a nonlinear cone. The Hasse argument is applied after filtering by the first ordinary total degree, so it bounds each exponent of every leading monomial. It is not the invalid inference that one low-digit-weight coefficient must survive an unknown linear change.

The rank descent always decomposes a particular target interval of generic rank r. That is what prevents a greedy prefix from jumping below r, and ensures every adjacent-block and tail test has rank exactly r. Conditioning on the outside coefficients leaves constant endpoints for the chosen chunk, even when those endpoints depend on all old seeds. The coefficient field in the generic Hasse relations remains the original field, where each shifted latent argument is transcendental. These are the interfaces on which the proof depends.

The final conclusion is an internal candidate proof with polynomial size for fixed width. It does not give a polynomial bound uniform in unbounded width, assert an optimal parameter count, or cover singular affine projections or orbit closures.

References

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